WORKING PAPER · FINE ART MEDIA & RMIT UNIVERSITY

ARCHAI

Sovereign, local-first AI for cultural heritage — turning collections into *living archives* people can talk to, while every record stays on hardware the institution owns and controls.

Research, technical overview & collaboration agenda Rob Graham · rob@fineartmedia.tech
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01 The research

Cultivating a Living Archive: Designing Sovereign AI Infrastructure for Cultural Heritage. A practice-based Design PhD at RMIT University, developed through Research through Design — where building the working system *is* the primary act of discovery, not an illustration of theory. A related paper was accepted for the 6th Summit on New Media Art Archiving at ISEA2026, Dubai.

Primary research question

How can design transform cultural heritage collections into living archives through sovereign, local-first AI infrastructure while preserving provenance, institutional authority and curatorial practice?

Secondary questions

1. What design principles support trustworthy conversational engagement with collections while maintaining archival integrity?
2. How can sovereign, local-first AI improve accessibility without compromising institutional control of cultural knowledge?
3. How might conversation strengthen public engagement while complementing, rather than replacing, curatorial expertise?
4. What transferable design framework could support ethical adoption of sovereign AI across museums, galleries, libraries and archives?

02 What ARCHAI is

Not a new collection system — a **semantic interface layer** that sits *above* whatever an institution already runs (EMu, Axiell, Vernon, TMS, CollectiveAccess, ResourceSpace, and the rest). It connects via their APIs, reads and normalises the metadata, and makes the collection **semantically searchable and conversationally accessible** — no migration, nothing replaced, and it never writes back to the record.

Because it runs on institutionally owned hardware with local-first models, the collection *and* the new layer of visitor conversation it creates stay in the building. Sovereignty here is not just an ethical preference — it is a preservation requirement: an institution that can't maintain independent access to its own interpretive systems has, in a real sense, lost control of its collection.

> **The living-archive proposition.** Instead of a static repository visitors must search, the archive becomes a place that can hold a conversation while staying anchored in evidence. Every response is bounded by verified collection information — the archive doesn't become less authoritative, its knowledge becomes more accessible.

03 Technical overview

Three-layer architecture

HERITAGE FOUNDATION	DERIVATIVE PROCESSING	EDITORIAL
The institution's CMS / DAMS	**Vectors + AI responses**	**Curator interpretation**
Read-only. No AI output ever enters the record. Permanent, authoritative, citable.	Entirely regenerable — rebuilt from unchanged source when models improve.	Flows one way, archive → visitor. Never writes back.

The stack — all on-device

SEARCH Qdrant + local embeddings	REASONING Local LLM (Qwen), five-layer grounding	VOICE Whisper + MeloTTS, multilingual
ACCESS CORS-locked Cloudflare tunnel		

Six technical contributions

- **Vector embeddings for heritage discovery** – semantic search without depending on controlled vocabulary.
- **Provenance-tracked interpretation** – every answer records its derivation chain back to source.
- **Layered architecture** – permanent heritage held apart from regenerable AI processing.
- **Conversational objects (AUXIO)** – items speaking in the first person from their own validated metadata.
- **Five-layer hallucination prevention** – responses stay grounded under adversarial conditions.
- **Transferable hardware reference** – a self-deployable stack for the GLAM sector.

Live today: ~1,400 public AUXIO pages across institutions on five continents; 3,000+ staff-searchable records from twenty open sources; every object open-licensed, rights-restricted media excluded at ingest. Deployment is a **one-time hardware capital cost, owned outright — not a per-seat cloud subscription.**

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04 A research agenda for the museum

Seven Research-through-Design experiments – each a working prototype tested in a real institution, mapping directly onto the questions above.

1 · Curatorial collection intelligence `STAFF-FACING`

How does conversational + semantic search change how curators find connections and build exhibitions?

Give Collections and Exhibitions staff ARCHAI's staff mode over a scoped set; measure new connections surfaced, time-to-find, and real use in exhibition development. The least-served, most-wanted, lowest-risk place to start.

2 · Accessibility experiments `VISITOR-FACING`

What access gains come from conversational, multilingual, voice-first interpretation?

Multilingual AUXIO, voice in/out, plain-language and reading-level adaptation, cognitive-load pacing, high-contrast low-vision modes. The strongest external-funding fit – it extends existing museum accessibility and digital-practice research.

3 · Local-LLM evaluation → the sovereign appliance `TECHNICAL`

Which local models sustain grounded, hallucination-free, fast interpretation on museum-owned hardware?

Benchmark local models on institutional records against the five-layer grounding framework – fidelity, latency, power – and publish a hardware reference specification.

4 · E-paper signage + NFC `PHYSICAL`

Can low-power e-paper + NFC turn a static label into living, updatable, conversational interpretation?

E-paper labels that redraw from the record (always-on, sustainable); NFC tap → an AUXIO conversation on the visitor's own phone, no app. Prototyped in-gallery.

5 · The federated sovereign index `FLAGSHIP`

What becomes possible when all open-licensed collections are federated into one local sovereign stack?

Scale the index into a "connecting archives" brain — cross-collection discovery, provenance across sources, sector-wide questions — held locally and owned outright. The largest ambition, and a natural ARC Linkage-scale project.

6 · The living archive `COMMUNITY PROVENANCE`

Can moderated visitor contributions become verified archival metadata without eroding authority?

A public message wall gated by an on-device safety layer (crisis routed to human help, hate held for review); verified facts earn a ★ **green star** and are added to the object's record by curators in the staff app. Community knowledge, feeding the archive.

7 · Cultural protocols as architecture `GOVERNANCE`

How can cultural protocols be enforceable properties of the system, not just written policy?

Co-design a cultural-sensitivity layer with the institution's First Nations governance — see the next section, which this experiment leads.

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05 First Nations collections & data sovereignty

Linking First Nations collections to *commercial cloud* AI is precisely what Indigenous Data Sovereignty frameworks forbid — it moves the material offshore and out of community control. A sovereign, local-first system is the opposite by design, and this is where ARCHAI's "ethics as architecture" thesis is tested most seriously.

The systems

- **Mukurtu CMS** — Indigenous-led, open-source, with differential access and Traditional Knowledge narratives in the metadata.

- **Traditional Knowledge (TK) Labels + Local Contexts Hub** – the protocol/rights layer; communities author the labels and control their delivery.
- **Ara Irititja** and **AIATSIS** – Australian community and national heritage systems.

How ARCHAI would engage — and the line it holds

ARCHAI can **read and enforce TK Labels**: exclude restricted material at ingest, honour differential access, foreground community narratives, and never write AI output back into the record. Running locally, the data never leaves – satisfying **Authority to Control** and infrastructure sovereignty (CARE Principles; Maiam nayri Wingara).

> This can only be a **community-governed partnership, consent-first**. Authority to Control sits with communities. Not everything is linkable, and some material must never be ingested or surfaced. ARCHAI's role is to *respect and enforce* protocols; communities decide. The proposition is not "ingest Indigenous collections" – it is "sovereign infrastructure that enforces *your* protocols, on *your* terms, on hardware you control."

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06 Working together

RMIT is already ACMI's Major University Partner – a decade-long relationship, renewed in 2026, spanning the ACMI + RMIT Audience Lab, the ARC-supported Museum Digital Social Futures project, RMIT's Centre of Digital Ecosystems, and an existing ARC Fellowship. ARCHAI extends that partnership into sovereign AI research. Pathways:

- **Scoped pilot** – a small, low-risk, ACMI-approved record set; rights, terminology and cultural protocols set by your team.
- **ARC Linkage Project** – RMIT + ACMI as partners; Major-Partner status strengthens the bid.
- **Audience Lab experiments** – in-gallery prototypes (e-paper/NFC, the living archive) tested with real audiences.
- **Visitor studies & evaluation**, **co-authored outputs**, and **student WIL projects** extending the existing pipeline.

What we'd explore together

? Which collection subset makes the best low-risk pilot?

- ? Which CMS / DAMS should ARCHAI read from, and what does the API expose?
- ? What rights, licensing and First Nations governance must be enforceable from day one?
- ? A light informal pilot, or a formal research partnership / ARC Linkage bid?
- ? Could the museum host visitor studies to evaluate the living-archive model?

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Try the concept (best on a phone): fineartmedia.tech/acmi

Sources on First Nations frameworks: Mukurtu CMS & TK Labels (mukurtu.org) · Local Contexts (localcontexts.org) · Ara Irititja (irititja.com) · CARE Principles (gida-global.org) · Maiam nayri Wingara (maiamnayriwingara.org). An independent proposal — not affiliated with or endorsed by any institution named.